

RITIKA KULKARNI

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PROFESSIONAL SUMMARY

Results-driven AUTOSAR Classic Platform BSW Developer with 3+ years of experience in Automotive Embedded Systems, specializing in Communication Stack development across ADAS USS-Based Parking and Commercial Vehicle (CV) product lines. Proficient in **CAN, CAN-FD, and J1939 communication protocols, AUTOSAR BSW stack configuration** (COM, PduR, RTE, CanIf, CanTp, NM, Dcm), **SWC design and interface mapping**, E2E Protection, and RDTs-based validation — with full SDLC exposure from requirements to deployment under **ASPICE and Agile** frameworks. Certified in **ISO 26262 Functional Safety, Automotive Ethernet / SOME-IP**, and **Oracle DevOps**; actively expanding expertise toward AUTOSAR Adaptive Platform (AP) and CI/CD-driven delivery pipelines.

CORE SKILLS & TECHNOLOGIES

AUTOSAR Classic BSW	COM, ComM, PduR, CanIf, CanTp, Nm, Dem, Dcm, NvM, E2E Protection, RTE, SWC Design & Integration
J1939 Stack	DBC Preparation & Integration, J1939Nm, J1939TP, J1939Dcm, PGN/SPN Analysis, Address Claiming, Signal Mapping
Functional Safety	ISO 26262 (ASIL concepts), MISRA C, Polyspace, QAC, OSS Analysis, ASPICE L2 process compliance
AUTOSAR Adaptive (AP)	Foundational knowledge of AP architecture, ara::com, SOME/IP, Automotive Ethernet — actively upskilling
Testing & Validation	CANoe, CANalyzer, Lauterbach Trace32, RDTs, SSM, GTest, Unit Testing, Code Coverage Analysis
Config & Dev Tools	EB Tresos (Elektrobit), Vector MICROSAR awareness, AEEE Pro, Git, Linux/WSL, Jira, Bitbucket, JFrog Artifactory, VS Code
CI/CD & Cloud	Jenkins (pipeline awareness), JFrog Artifactory, AWS EC2 & S3, Oracle Cloud DevOps
Requirements & ALM	IBM DOORS, Codebeamer, RQM, Splunk; full traceability from DOORS requirements to test artifacts
Programming	C (primary), C++, Python; AUTOSAR OS / RTOS concepts

PROFESSIONAL EXPERIENCE

Platform Engineer – BSW | Bosch Global Software Technologies (BSW) | Bengaluru, India | July 2023 – Present

Project 1: MPC4-Based Commercial Vehicle (Truck) Platform – AUTOSAR BSW Feb 2026 – Present

Domain: J1939 Communication | Stack: AUTOSAR Classic | Tools: EB Tresos, CANoe, CANalyzer, IBM DOORS

- Analyzed Public CAN and Private CAN network matrices for a commercial truck platform; authored J1939-compliant COM DBCs covering **40+ PGNs**, reducing rework cycles by eliminating signal mapping ambiguities early in the integration phase.
- Configured and integrated **J1939TP, J1939Dcm, and J1939Nm** modules in EB Tresos, achieving full protocol compliance validated via CANoe test sessions — zero Sev-1 defects post-integration.
- Performed **PGN/SPN analysis and CAN signal mapping** for vehicle communication architecture; documented Address Claiming and Source Address allocation per J1939-81, enabling plug-and-play node management across 3 ECU variants.
- Designed and mapped **SWC (VehicleInput / VehicleOutput) application interfaces** through DADDY AUTOSAR toolchain, reducing interface integration effort by standardizing port connector definitions.
- Supported **Vehicle Generic Interface (VGI)** mapping by cross-referencing signal definitions in IBM DOORS against DBC configurations, maintaining 100% traceability between requirements and implementation artifacts.
- Leveraged AI-assisted engineering tools for J1939 DBC analysis, interface mapping, and technical documentation — cutting configuration review time by ~30%.

Project 2: AUTOSAR BSW – Volkswagen USS-Based Parking System (EVO_MBQBL & MEB_UNECE) July 2023 – Jan 2026

Domain: ADAS – Ultrasonic Sensor Parking Assist | Stack: AUTOSAR Classic | Tools: EB Tresos, CANoe, Trace32, GTest, RDTs, Polyspace, QAC

- **BSW Stack Configuration & RTE Integration:** Configured COM, ComM, PduR, CanIf, CanTp, NM, Dem, Dcm, NvM, and RTE for ADAS ECU — enabling reliable signal routing across Application, RTE, and COM layers with **zero critical signal-loss defects** reported in system integration.
- **SWC Design & Signal Routing:** Authored and validated **SWC port interfaces and DBC-based TX/RX message mappings**; implemented signal conversion functions and confirmed end-to-end flow via CANoe traces — covering **60+ CAN signals** across two product variants (EVO and MEB).
- **Defect Resolution (ASIL-aware):** Resolved **25+ Change Requests** and internal BSW defects by analyzing Lauterbach Trace32 logs and COM-layer traces; root-cause analysis improved first-time-fix rate from 60% to 85% over two release cycles.
- **Testing & Code Coverage:** Executed SSM, GTest, and unit test suites achieving **>90% MC/DC code coverage** on communication stack modules; authored test reports traceable to IBM DOORS requirements through RQM.
- **Static Analysis & MISRA C Compliance:** Resolved **100+ Polyspace and QAC findings** (including MISRA C:2012 mandatory violations) across two software drops; maintained OSS license compliance through structured review.
- **ASPICE L2 Process Compliance:** Maintained full artifact traceability from DOORS requirements through implementation and test evidence in RQM — supporting successful ASPICE audit with **zero open process non-conformances**.

EDUCATION

B.E. in Telecommunications Engineering | Bangalore Institute of Technology, Bengaluru | 2019 – 2023 | CGPA: 8.65 / 10

- **2nd Place – INCEPTRA 2023** National Level Project Exhibition: Designed *iBuzz Driver Safety Monitoring System* using IoT sensors and real-time alerting — relevant to ADAS domain expertise.
- Newsletter Designer, TESA (Telecommunications Engineering Students' Association).

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified DevOps Professional
- AUTOSAR Classic Platform Fundamentals
- Automotive SPICE (ASPICE) Fundamentals – Process Ref. Model
- ISO 26262 Functional Safety Fundamentals (ASIL A–D)
- CAN, CAN-FD & J1939 Communication Protocols
- Automotive Ethernet & SOME/IP Fundamentals

ADDITIONAL INFORMATION

Languages: English (Professional) | Kannada (Native) | Hindi (Conversational) **Date of Birth:** 14 March 2001 **Hobbies:** Event Emceeing / MC, Cooking, Art & Dance, Fun@Work Team Member